

Distributed and Dynamic AI Agent Collaboration over Optical Transport for Network Orchestration and Monitoring

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Abstract: We demonstrate a dynamic multi-agent system distributed over 811-km optical transport networks for cross-domain orchestration and monitoring. By selecting optimal agents based on instantaneous conditions, the system achieves greater stability and $2.9\times$ lower latency. © 2026 The Author(s)

1. Introduction

AI agents are autonomous systems designed to perceive their environment, make decisions, and act toward achieving specific goals. Distributed agents, which operate collaboratively across networked environments, can leverage resources across the entire network and be shared among multiple end users, enhancing scalability, efficiency, and adaptability. However, distributed agents will be highly sensitive to their operational contexts. Network-level factors such as port speeds and fiber link latency, along with compute-level factors such as CPU and GPU utilization, directly influence agent performance. The dynamic nature of these conditions makes it challenging to sustain optimal agent performance, which in turn affects collaboration in multi-agent systems. This work investigates a dynamic agent selection strategy to overcome this instability. By continuously evaluating the state of a homogeneous agent pool and routing requests accordingly, our approach aims to evade resource contention, thereby maintaining high levels of agent collaboration and overall operational efficiency.

We instantiate the selection strategy through a highly capable distributed multi-agent [1] system, in which specialized agents—powered by embedding models, large language models (LLMs), visual language models (VLMs) [2], and audio language models (ALMs) [3]—collaboratively perform complex network management tasks. Their functionalities span fiber health monitoring [4], real-time analysis of surveillance feeds, IP service orchestration [5], and overall network management [6, 7]. These agents are deployed as redundant instances interconnected by a low-latency optical transport network. Crucially, by integrating real-time operational monitoring, the system can dynamically select the most suitable agents of a given type to participate in any collaboration, forming adaptive agent groups. Experimental results demonstrate that this approach to distributed agent collaboration provides significant improvements in operational efficiency in a dynamically changing environment.

2. Dynamic Multi-Agent Collaboration

Figure 1(a) illustrates the architecture for dynamically forming a multi-agent group by scoring and selecting (S&S) the most suitable agent from six subgroups, each composed of identical specialized agents. All the agents are actively running to avoid startup delays. To identify the optimal agent, multiple operational factors are evaluated, including the model's average inference latency, the data transmission rate of the associated network port, GPU

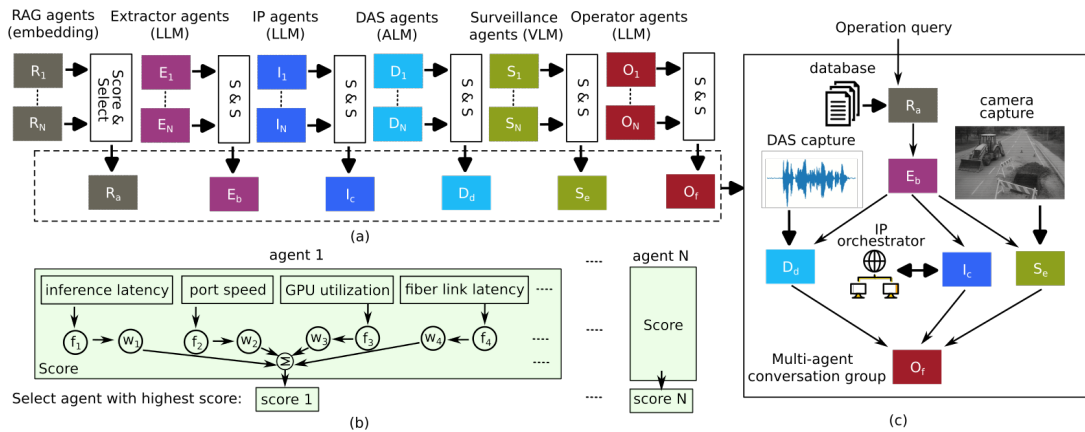


Fig. 1: (a) Dynamic formation of an AI agent group by scoring and selecting (S&S) the optimal agent from each specialized subgroup; (b) a detailed view of the score-and-select process and (c) diagram of a multi-agent conversation group for network orchestration and monitoring.

memory utilization, fiber link latency, and others, as shown in Fig. 1(b). Each factor is assigned a weight w_x in the scoring function f_x , and the weighted sum yields a final score, with a higher score indicating that the agent is in the most favorable state for selection compared to its identical peers.

The six selected agents then collaborate to perform fiber monitoring using distributed acoustic sensing (DAS) and camera feeds, while also providing IP connectivity status, as shown in Fig. 1(c). We employ the open-source Qwen model family running on lab servers to ensure data security, and all agents exchange messages with one another using gRPC. The Retrieval-augmented Generation (RAG) agent (R_a), powered by Qwen3-Embedding-4B, receives natural-language requests from network operators and retrieves the most relevant document chunks—such as from product manuals—from a database pool. The Extractor agent (E_b), built on Qwen2.5-7B-Instruct, then extracts the necessary node parameters, including IP addresses, system identifiers, interface details, and connected router information to support parallel monitoring of DAS, surveillance, and IP status. For fast and accurate interpretation of DAS audio signals, we fine-tuned Qwen2-Audio-7B-Instruct model using 271 DAS audio samples [8] labeled as either construction equipment sound or background noise. The dataset was split into 80% training and 20% evaluation sets, and fine-tuning was performed with LoRA [9] using a learning rate of $2e-5$. After six training epochs, the model achieved 100% evaluation accuracy. In deployment, the DAS agent (D_d) with this fine-tuned model processes DAS sensing data every 10 seconds, enabling the DAS agent to determine whether a signal indicates construction activity. The IP-domain agent (I_c), also powered by Qwen2.5-7B-Instruct, uses the extracted information to query associated tools and verify the operational status of multiple layers of IP. The Surveillance agent (S_e) uses Qwen2.5-VL-7B-Instruct to identify the surveillance area based on the extracted optical path, retrieves images of the fiber region, and detects construction activity from the images. Finally, by aggregating the outputs from the DAS, IP-domain, and Surveillance agents, the Operation agent (O_f) determines the necessary next steps, supporting subsequent actions such as building connectivity between servers.

3. Experimental Demonstration

The effectiveness of dynamically forming distributed agents for comprehensive task solving is demonstrated on a real optical transport network comprising five Nokia 1830 Photonic Service Switch (PSS) nodes, as shown in Fig. 2(a). The topology includes three compute clusters (C_1 , C_2 , and C_3) interconnected by 400-G IP routers. Cluster C_1 is optically linked to C_2 over 288 km and to C_3 over 468 km of spooled fiber. C_1 contains three servers, each equipped with a GPU for inference. Agents E_1 , I_1 , and O_1 are co-located on a single server, allowing them to share the same LLM since they are executed sequentially within the multi-agent process. This configuration is mirrored by agents E_2 , I_2 , and O_2 , allowing redundant agents of each type to compete with one another. Clusters C_2 and C_3 each host a server with two GPUs, enabling parallel execution of two agents (D_1 , S_1 and D_2 , S_2) per server and fostering competition between identical agents. Agents on a given server share the same physical Ethernet port and therefore experience identical network traffic conditions. To exemplify the impact of operational factors, we measure group conversation duration under controlled stress. This includes imposing heavy GPU utilization on all unique agents except R_1 , and applying severe network rate-limiting to the servers for S_1/D_1 and $E_1/I_1/O_1$ to simulate network congestion. Fig. 2(b) shows that, compared with the normal case under light load and minimal traffic shaping (highlighted in green as the baseline), GPU overload on the Surveillance agent S_1 increases the group conversation duration by $2.5\times$, while network rate limiting below 1 Mbit/s on the server hosting S_1/D_1 leads

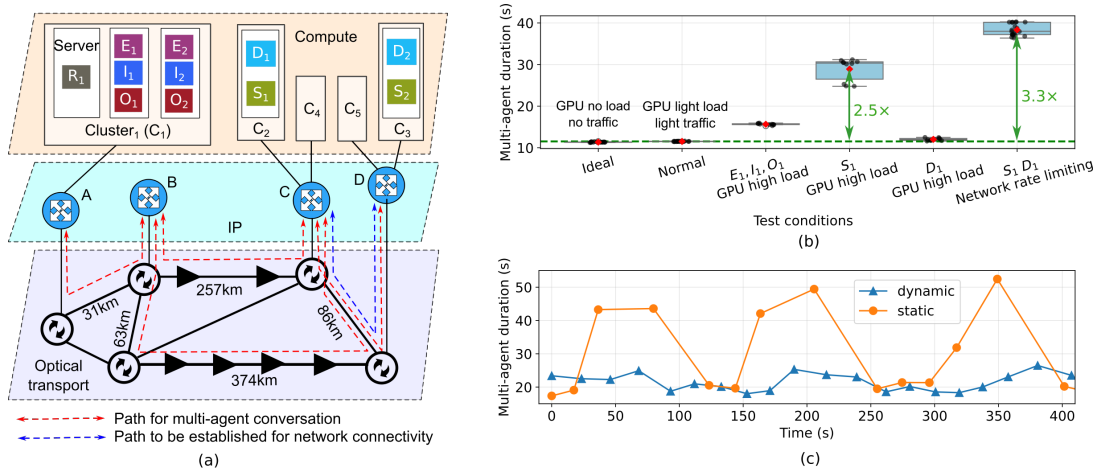


Fig. 2: (a) Experimental topology for distributed AI agent collaboration over an optical transport network. (b) Measured multi-agent conversation durations under different conditions; red points show means, green line indicates the normal-condition mean. (c) Comparison of conversation durations for dynamic versus static multi-agent approaches, with and without optimal agent selection, respectively.

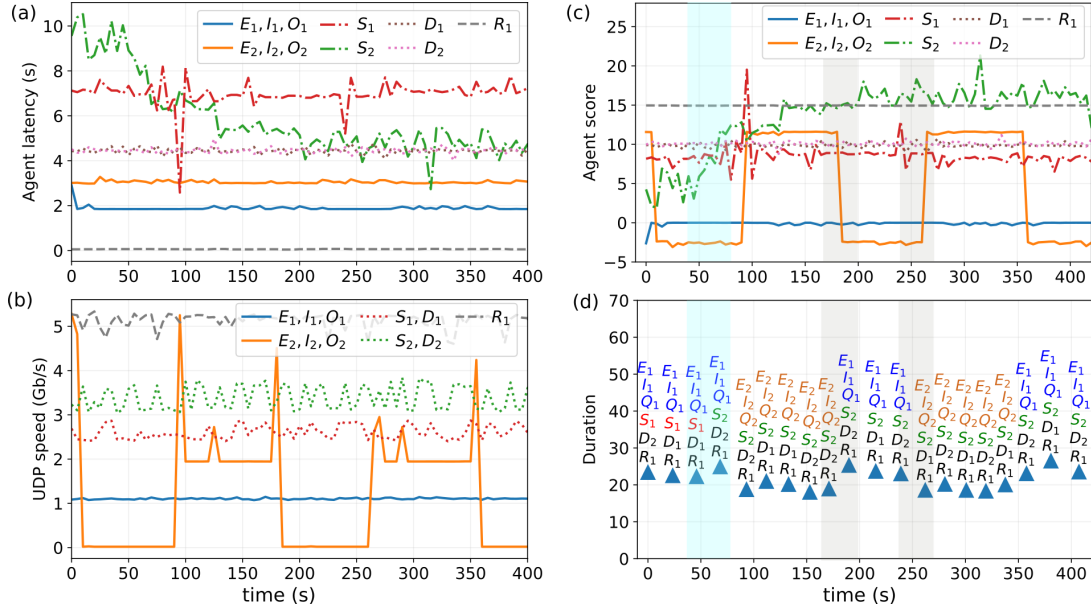


Fig. 3: (a) End-to-end inference latency and (b) UDP throughput for each agent over 400 seconds with a 5-second update interval; (c) computed agent scores based on monitored parameters; (d) durations of the dynamic multi-agent process with optimal agent updates.

to a 3.3× increase or even service failure due to inference timeouts on $E_1/I_1/O_1$. Thus, when no agent switching occurs and the GPU load becomes high and/or network traffic becomes constrained, the multi-agent processing time increases significantly. In contrast, by continuously selecting the optimal agent and dynamically forming a rational multi-agent group in response to performance changes, their impact on processing time can be effectively limited, as shown in Fig. 2(c).

To demonstrate dynamic agent collaboration, we test with a network where inference latency and port speed vary over time and are monitored in real time. Fig. 3(a) and (b) present the average end-to-end request latency per second for each agent and the maximum achievable UDP bandwidth of a single server, respectively. S_2 's latency decreases over time and eventually intersects with that of S_1 in Fig. 3(a), while their traffic limits in Fig. 3(b) show only minor fluctuations without crossing. The server hosting agents E_2, I_2 and O_2 shows periodic variations in UDP bandwidth in Fig. 3(b), yet their end-to-end latencies remain relatively stable in Fig. 3(a). Scores for each agent are computed such that average latency contributes linearly, while changes in UDP bandwidth are periodic and binary, as illustrated in Fig. 3(c). Fig. 3(d) shows the multi-agent execution timeline with agent switching. As S_2 's score surpasses S_1 's in Fig. 3(c), agent S_2 replaces S_1 in the group conversation, as indicated by the blue-shaded region in the plot. Likewise, the gray-shaded region shows that when the scores of E_2, I_2 and O_2 drop below those of E_1, I_1 and O_1 in Fig. 3(c), E_1, I_1 and O_1 are selected as the optimal agents. Conversely, when their scores exceed those of E_1, I_1 and O_1 , E_2, I_2 and O_2 become the selected agents. By dynamically selecting the optimal agents, the multi-agent group exhibits shorter and more uniform durations, achieving up to 2.9× acceleration compared with the higher and more variable durations observed when the agents remain static (see Fig. 2(c)).

4. Conclusion

While distributed multi-agent systems enhance accessibility, their performance is unstable under load due to disparate network and computing conditions. We address this with a dynamic agent selection strategy that ensures optimal performance by dynamically selecting the most suitable agents based on real-time operational conditions. In experiments on optical transport networks, our approach outperformed static configurations with shorter, more consistent task durations, highlighting the necessity of environment-aware orchestration for robust network management.

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